

CG2271 Realtime Operating Systems

Tutorial 3 Process Scheduling

1. [Putting it together] Take a look at the given mysterious program **Behavior.c**. This program takes in one integer command line argument **D**, which is used as a **delay** to control the amount of computation work done in the program. For the part (a) and (b), use ideas you have learned from **Lecture 3: Process Scheduling** to explain the program behavior.
 - a. **D** = 1.
 - b. **D** = 100,000,000 (note: don't type in the ", " ☺)
 - c. Now, find the **smallest D** that gives you the following interleaving output pattern:

Interleaving Output Pattern
<pre>[6183]: Step 0 [6184]: Step 0 [6183]: Step 1 [6184]: Step 1 [6183]: Step 2 [6184]: Step 2 [6183]: Step 3 [6184]: Step 3 [6183]: Step 4 [6184]: Step 4 [6184] Child Done! [6183] Parent Done!</pre>

What do you think "**D**" represents?

*Note: "**D**" is machine dependent, you may get very different value from your friends'.*

2. [Walking through Scheduling Algorithms] Consider the following execution scenario:

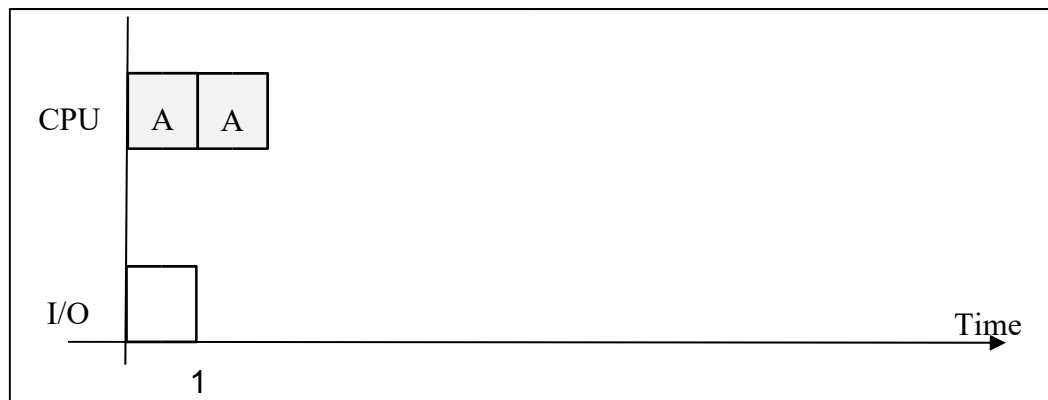
Program A, Arrives at time 0
Behavior (CX = Computer for X Time Units, IOX = I/O for X Time Units): C3, IO1, C3, IO1

Program B, Arrives at time 0
Behavior: C1, IO2, C1, IO2 C2, IO1

Program C, Arrives at time 3
Behavior: C2

- a. Show the scheduling time chart with First-Come-First-Serve algorithm. For simplicity, we assume all tasks block on the same I/O resource.

Below is a sample sketch up to time 1:



- b. What are the turnaround time and the waiting time for program A, B and C? In this case, waiting time includes all time units where the program is ready for execution but could not get the CPU.
- c. Use **Round Robin** algorithm to schedule the same set of tasks. Assume time quantum of **2 time units**.

- d. What is the response time for tasks A, B and C? In this case, we define response time as the time difference between the arrival time and the first time when the task receives CPU time.

3. We have the following set of tasks:

Task	C_i	P_i
T1	3	9
T2	5	18
T3	4	12

Is there an assignment of priorities to these 3 tasks so that they are schedulable?

4. Given a set of tasks with the following specifications:

Task	C_i	P_i
T1	2	6
T2	3	8
T3	4	15

- a) What is the CPU utilization of this set of tasks?
- b) Is this set of tasks RMS schedulable under the Liu and Leyland bounds?
- c) Schedule the tasks using RMS